Toward Hybrid H/V and Beamforming Site-Response Mapping

Figure 1. H/V spectral ratio overlay at station ANMO. The peak at 0.5 Hz indicates a sediment-basement interface.

In this study, we revisit the 2024 Sichuan sequence using a hybrid H/V and array-beamforming workflow. Additional context follows here for prose density.

We analyse data from 42 broadband stations deployed across the back-arc between January and August 2024. Additional context follows here for prose density.

The Mw 7.1 mainshock ruptured on 2024-04-03 at 22:14 UTC, triggering a 6-week aftershock sequence. Additional context follows here for prose density.

Our results recover the input velocity model to within 5% RMS at all depths shallower than 20 km. Additional context follows here for prose density.

Station Lat Lon Depth
ANMO 34.95 -106.46 1.8
CCM 38.06 -91.24 0.2
TUC 32.31 -110.78 0.9

The deployment geometry was optimised for back-azimuth resolution rather than depth resolution. We accept the trade-off because the study targets surface waves.

3. RESULTS

The hybrid workflow recovers Vs30 across the back-arc to within
12% of the borehole reference at 90% of co-located sites. Residuals correlate with surface geology, not station spacing.

We compare our method to two efficient baselines: a pure-H/V inversion and a pure-beamforming workflow. Our hybrid outperforms both above 0.3 Hz.

References

[1] Smith, J., 2022. H/V site response in the Cascadia back-arc. Bulletin of the Seismological Society of America, 112, 1234-1250.

[2] Wang, L., 2024. Array beamforming with sparse arrays. Geophysical Journal International, 238, 456-478.